

Homework (Carolina Reaper)

- Q1.** Find the slope of a line parallel to $y = 3x + 2$.
 (A) $m = 3$
 (B) $m = -3$
 (C) $m = 2$
 (D) $m = 1$
 (E) $m = 0$
- Q2.** A rhythm pattern has a frequency of $5 \cdot 10^2$ Hz. If a parallel pattern has the same slope, what is its frequency?
 (A) $5 \cdot 10^2$ Hz
 (B) 10^3 Hz
 (C) $2 \cdot 10^2$ Hz
 (D) 10^2 Hz
 (E) $5 \cdot 10^3$ Hz
- Q3.** What is the equation of a line parallel to $y = 2x - 1$ and passing through $(1, 3)$?
 (A) $y = 2x + 1$
 (B) $y = 2x - 1$
 (C) $y = x + 2$
 (D) $y = x - 2$
 (E) $y = 3x - 2$
- Q4.** Two parallel lines have slopes of m_1 and m_2 . If $m_1 = 2$, what is m_2 ?
 (A) $m_2 = 2$
 (B) $m_2 = -2$
 (C) $m_2 = 1$
 (D) $m_2 = -1$
 (E) $m_2 = 0$
- Q5.** A canvas has a width of 15 inches and a length of 20 inches. If a parallel line to the length has a slope of m , what is m ?
- (A) $m = 0$
 (B) $m = 1$
 (C) $m = \frac{4}{3}$
 (D) $m = \frac{3}{4}$
 (E) $m = -1$
- Q6.** What is the relationship between two parallel lines with equations $y = 2x + 1$ and $y = 2x - 1$?
 (A) They have the same y -intercept.
 (B) They have the same slope.
 (C) They have the same x -intercept.
 (D) They are perpendicular.
 (E) They have different slopes.
- Q7.** Two parallel lines intersect a line with equation $y = -2x + 1$. If the first line has equation $y = 2x - 1$, what is the equation of the second line?
 (A) $y = 2x + 1$
 (B) $y = 2x - 2$
 (C) $y = -2x + 1$
 (D) $y = x - 1$
 (E) $y = x + 1$
- Q8.** A song has a rhythm pattern with a frequency of $2 \cdot 10^3$ Hz and a parallel pattern with the same slope. What is the equation of the line representing the parallel pattern?
 (A) $y = 2x + 1$
 (B) $y = 2x - 1$
 (C) $y = x + 2$
 (D) $y = x - 2$
 (E) $y = 3x - 2$

Open Ended Questions (FRQ)

- Q9.** Prove that two lines with slopes m_1 and m_2 are parallel if and only if $m_1 = m_2$. Use a musical instrument as an example.
- Q10.** A painter creates a canvas with a width of 20 inches and a length of 30 inches. A parallel line to the length has a slope of m . Find the equation of the line and explain how it relates to the rhythm pattern of a song.

Chef's Tips

- Tip 1: Use unique scenarios and numerical values for each question.
- Tip 2: Ensure all math is wrapped in ... and use for LaTeX commands.
- Tip 3: Keep questions and options concise to fit on one page.